

Exp No. 6 :5G NR Frame Structure Analysis with Different Subcarrier Spacing

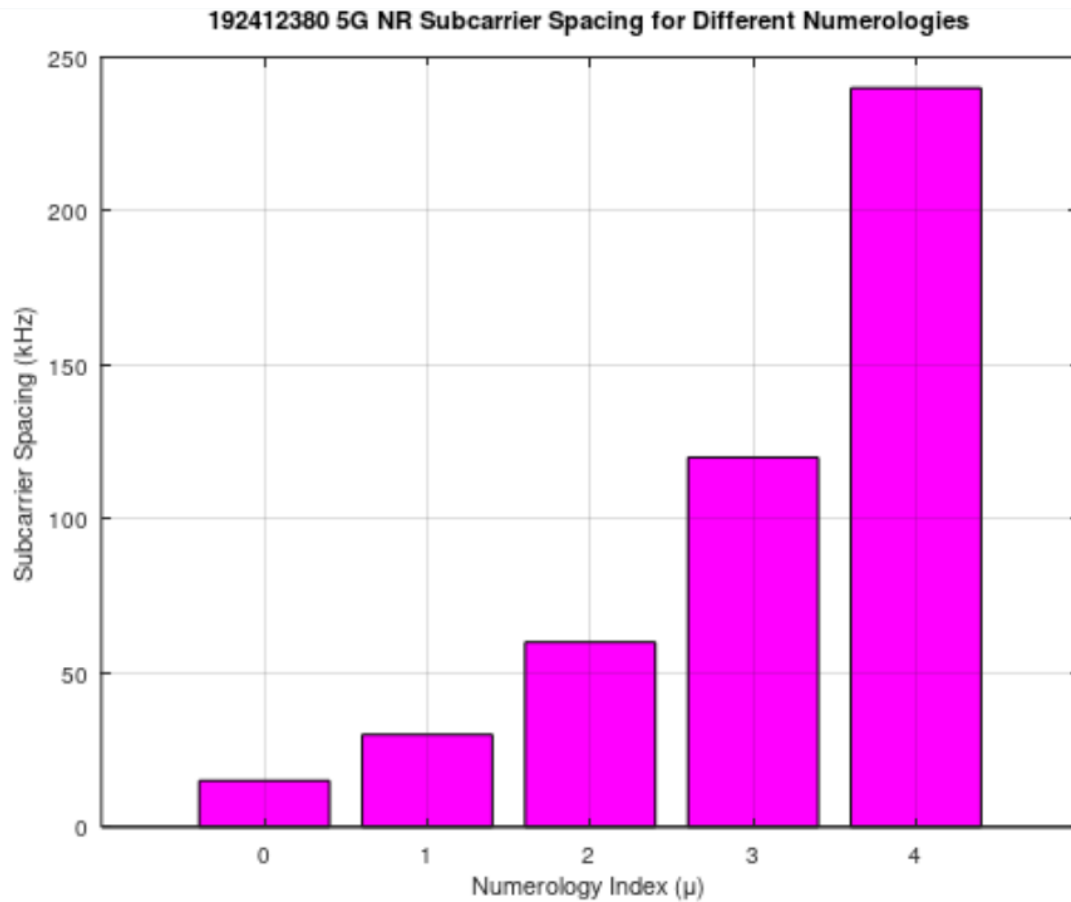
AIM: To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objective 1: To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % 5G NR Subcarrier Spacing (kHz)
6  scs = [15 30 60 120 240];
7
8  % Numerology Index
9  mu = 0:4;
10
11 figure;
12
13 % Bar Graph
14 bar(scs, 'm');    % Magenta bars
15
16 grid on;
17
18 set(gca, 'XTick', 1:5);
19 set(gca, 'XTickLabel', mu);
20
21 xlabel('Numerology Index (\mu)');
22 ylabel('Subcarrier Spacing (kHz)');
23 title('192412380 5G NR Subcarrier Spacing for Different Numerologies');
24
25 drawnow;
26 pause(5);
```

OUTPUT :



Objective 2: To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.

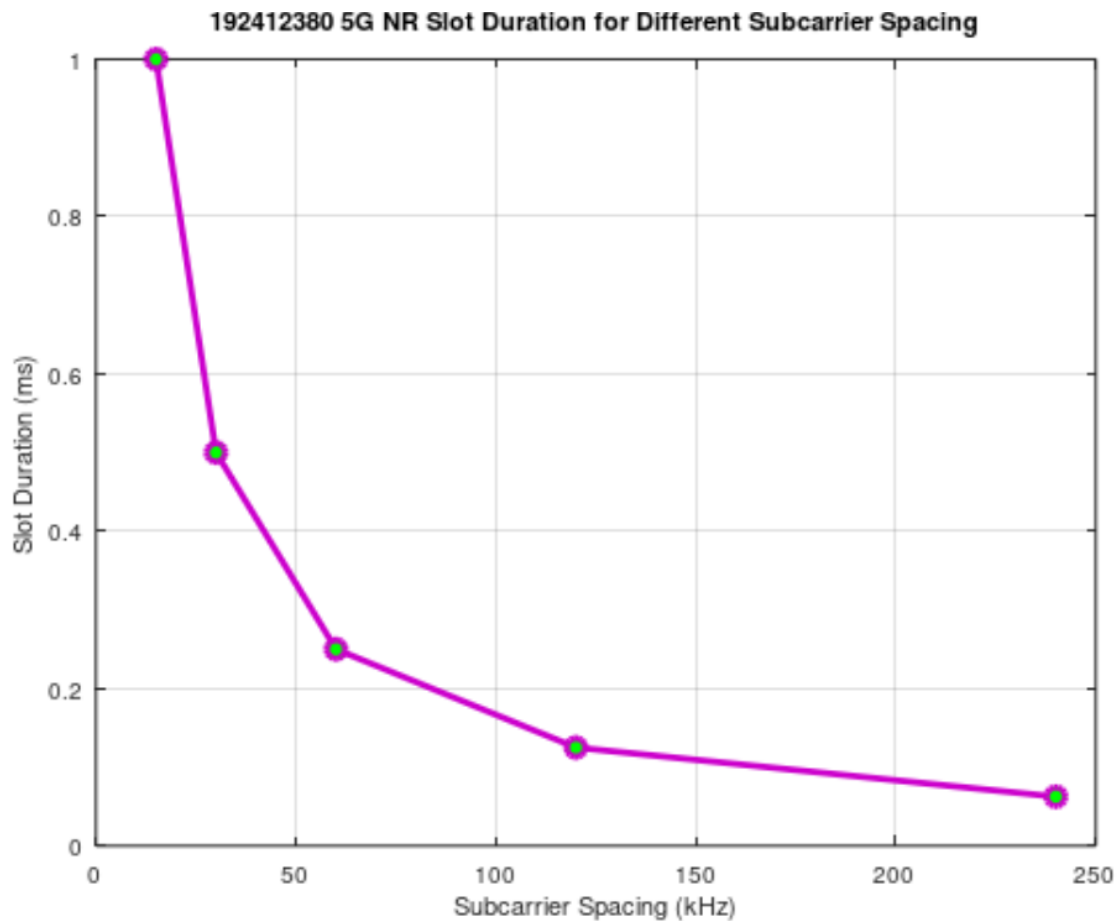
Program

```

1  clc;
2  clear;
3  close all;
4
5  % 5G NR Parameters
6  scs = [15 30 60 120 240];      % Subcarrier Spacing (kHz)
7  slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
8
9  figure;
10
11  plot(scs, slot, ...
12       '-o', ...
13       'Color', [0.8 0 0.8], ... % Purple line
14       'MarkerFaceColor', [0 1 0], ... % Green markers
15       'Linewidth', 2);
16
17  grid on;
18
19  xlabel('Subcarrier Spacing (kHz)');
20  ylabel('Slot Duration (ms)');
21  title('192412380 5G NR Slot Duration for Different Subcarrier Spacing');
22
23  drawnow;
24  pause(5);

```

OUTPUT :

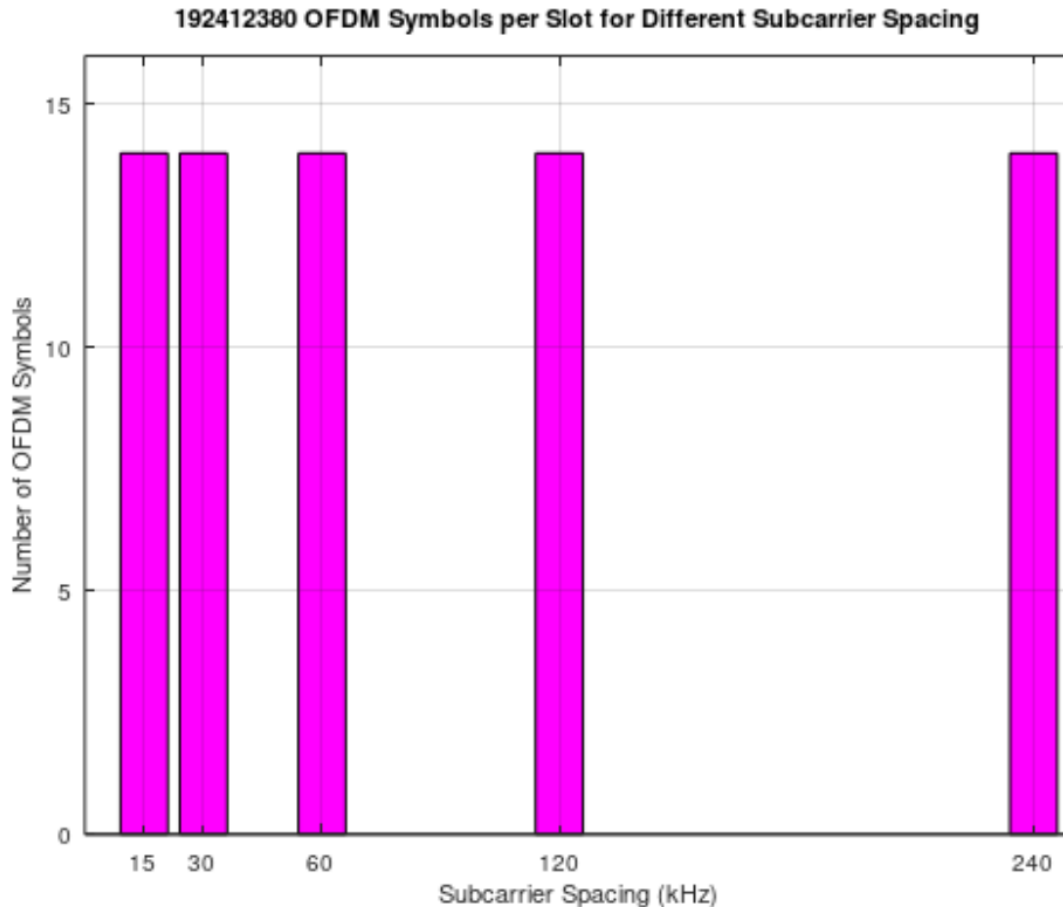


Objective 3: To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

program

```
1  clc;
2  clear;
3  close all;
4
5  % 5G NR Parameters
6  scs = [15 30 60 120 240];    % Subcarrier Spacing (kHz)
7  symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
8
9  figure;
10
11 % Bar Graph (Magenta Color)
12 bar(scs, symbols, 'm');
13
14 grid on;
15
16 xlabel('Subcarrier Spacing (kHz)');
17 ylabel('Number of OFDM Symbols');
18 title('192412380 OFDM Symbols per Slot for Different Subcarrier Spacing');
19
20 ylim([0 16]);
21
22 drawnow;
23 pause(5);
```

OUTPUT :



Result:

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.

Practical Questions

A 5G wireless communication system employs MIMO beamforming to improve signal directionality and system performance by steering the transmitted beam toward specific user locations. The steering angle of the antenna array plays a key role in determining beam direction, signal strength, and accuracy. Using MATLAB, simulate a MIMO beamforming system and analyze its performance.

1. To analyze the steering angle variation in a MIMO beamforming system and determine how the beam direction changes with different steering angles.
2. To compare the array gain achieved at different steering angles in a MIMO beamforming system and evaluate how beamforming improves signal strength.

3. To evaluate the beam steering accuracy by calculating the beam direction error between the desired steering angle and the actual beam direction.